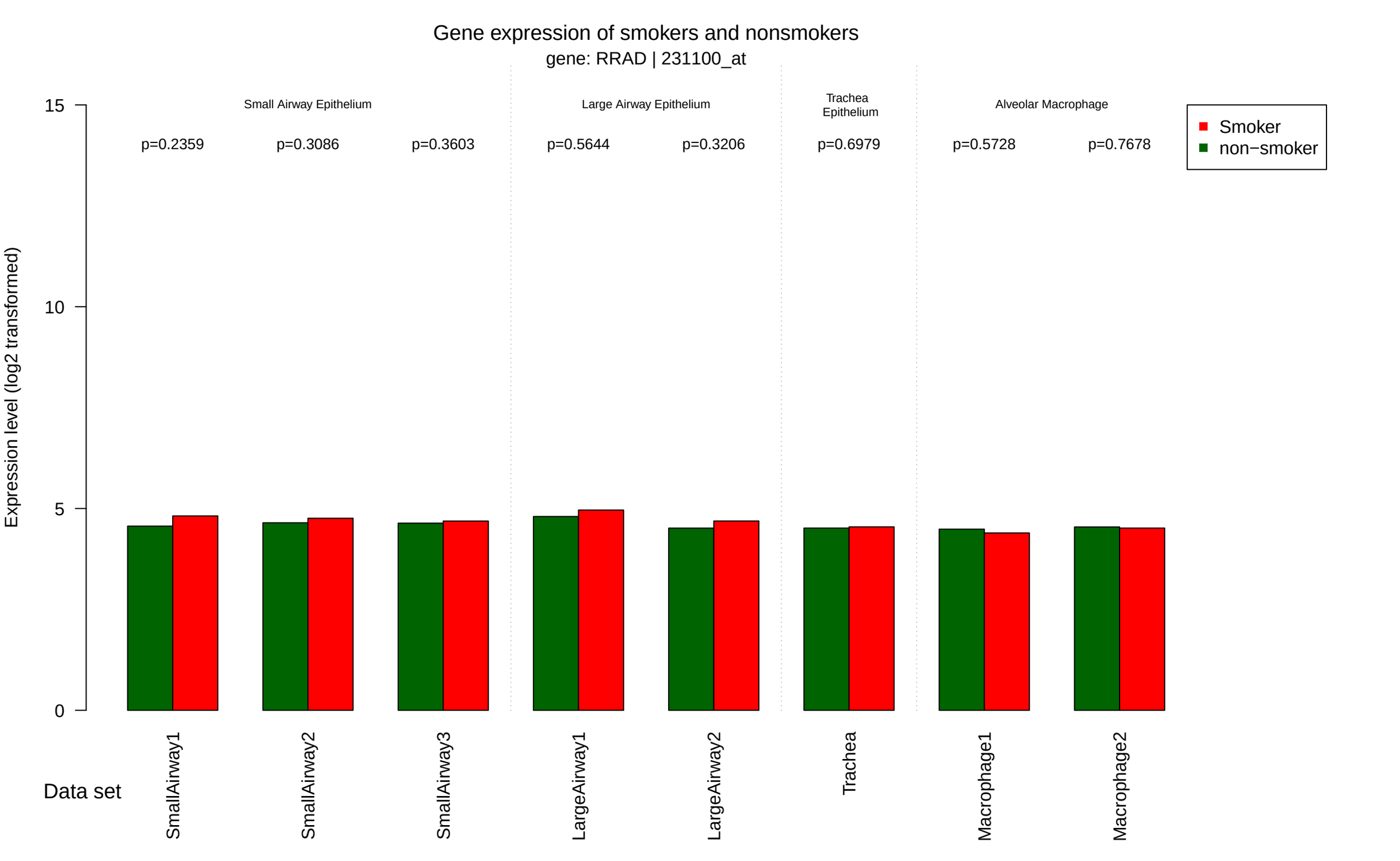
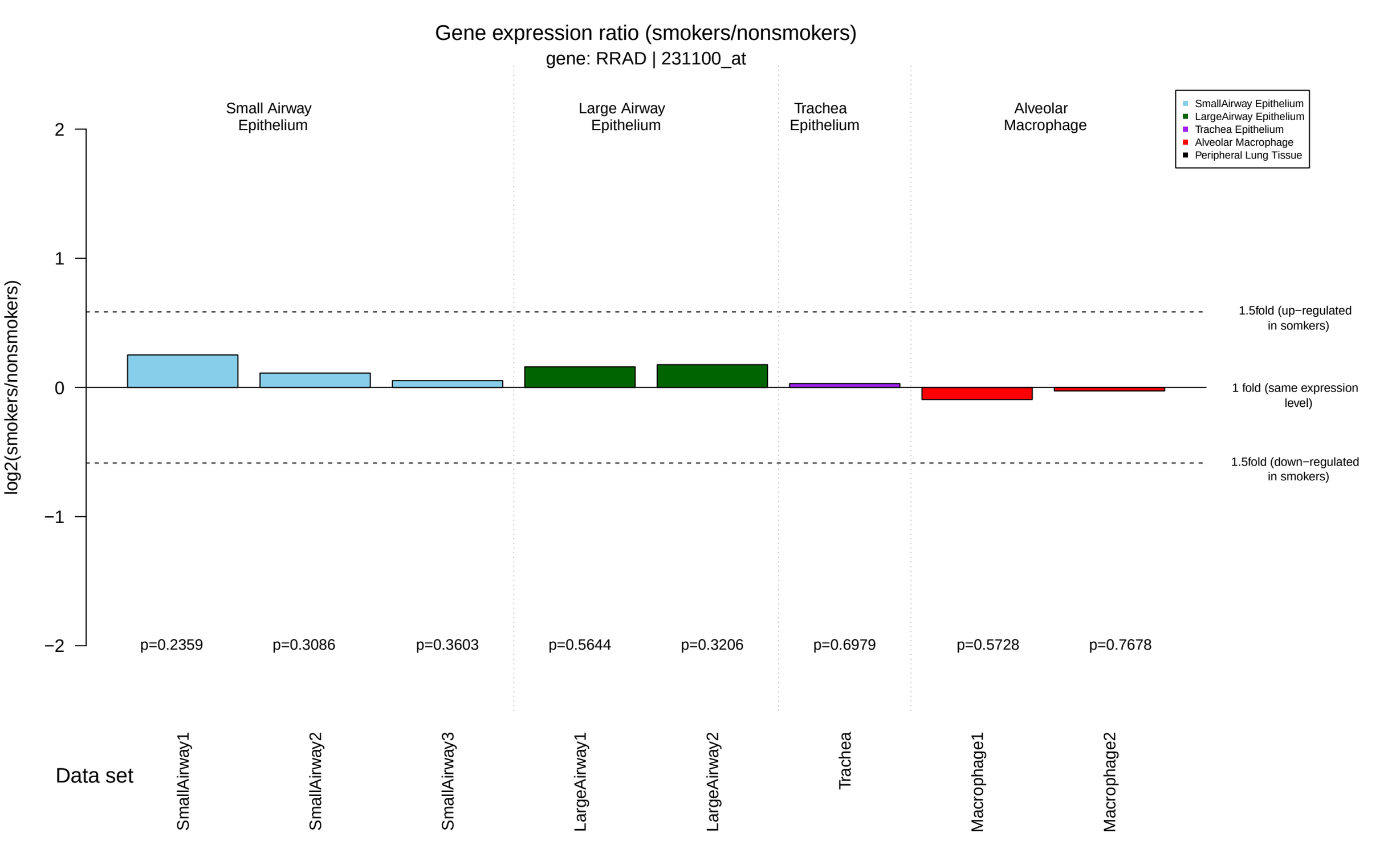
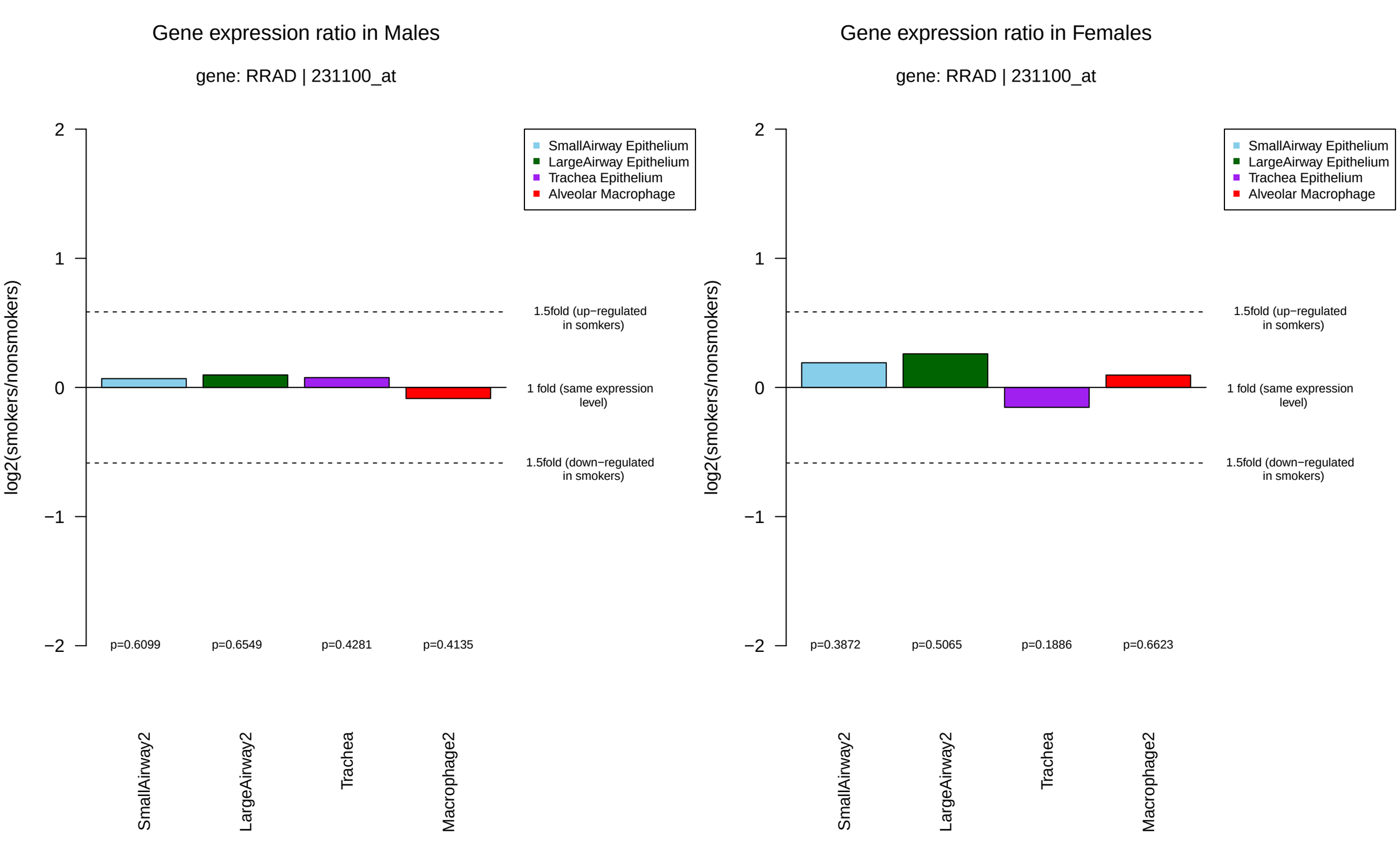
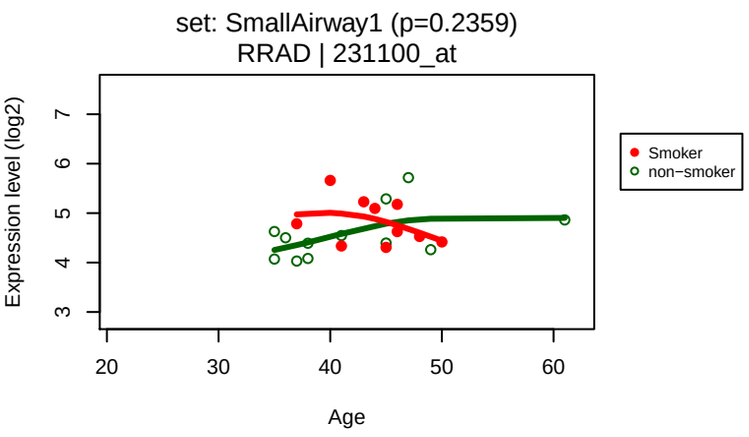
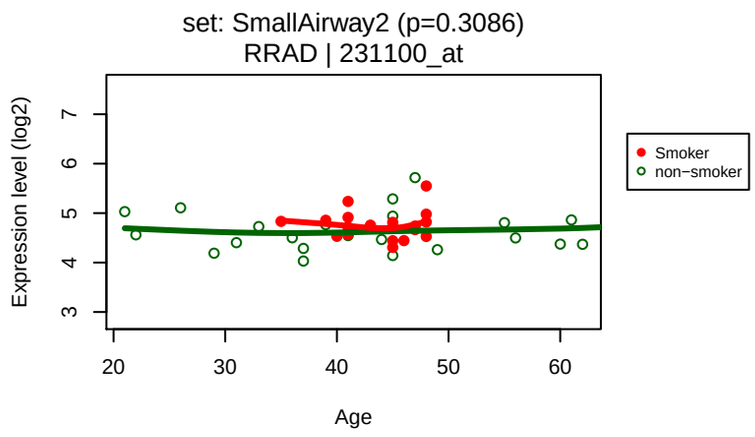
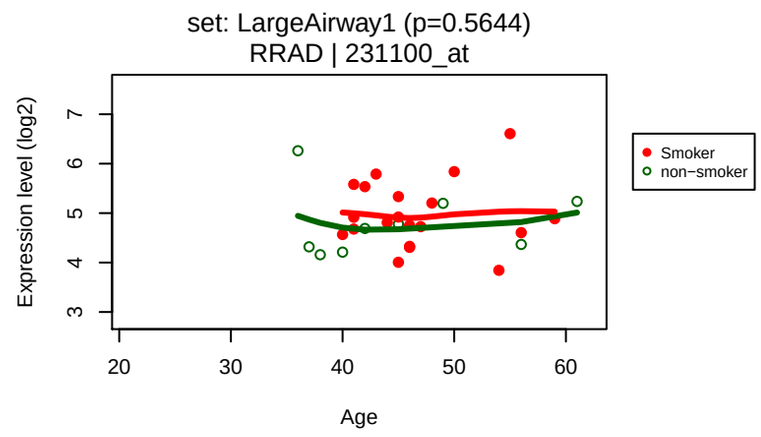

Gene expression of smokers and nonsmokers
gene: RRAD | 231100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2359
p=0.3086
p=0.3603
p=0.5644
p=0.3206
p=0.6979
p=0.5728
p=0.7678
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

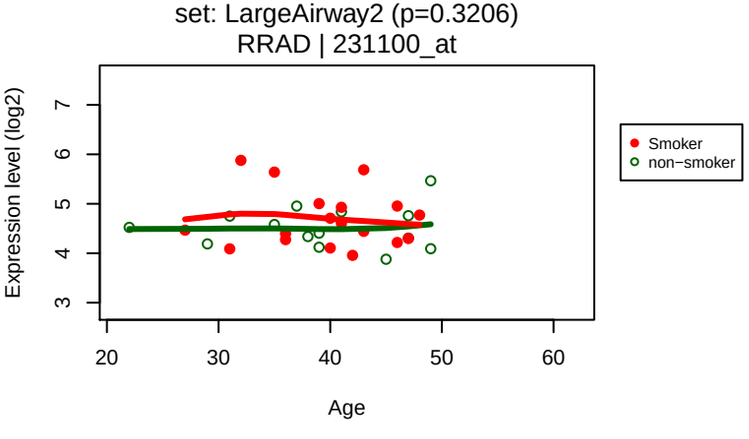

Gene expression ratio (smokers/nonsmokers)
gene: RRAD | 231100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2359
p=0.3086
p=0.3603
p=0.5644
p=0.3206
p=0.6979
p=0.5728
p=0.7678
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

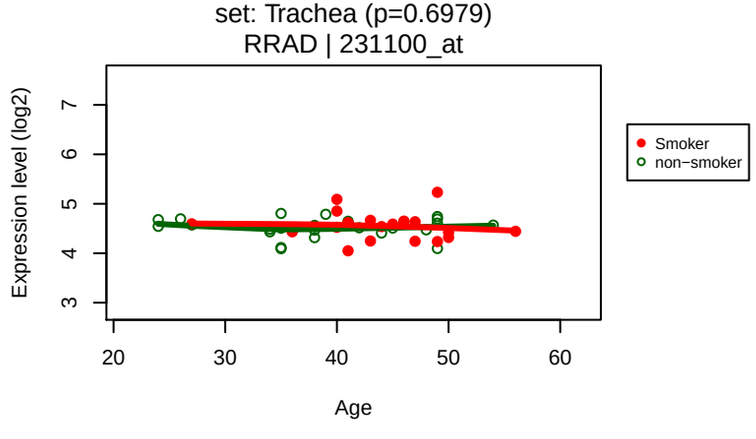

Gene expression ratio in Males
gene: RRAD | 231100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6099
p=0.6549
p=0.4281
p=0.4135
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRAD | 231100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3872
p=0.5065
p=0.1886
p=0.6623
SmallAirway2
LargeAirway2
Trachea
Macrophage2

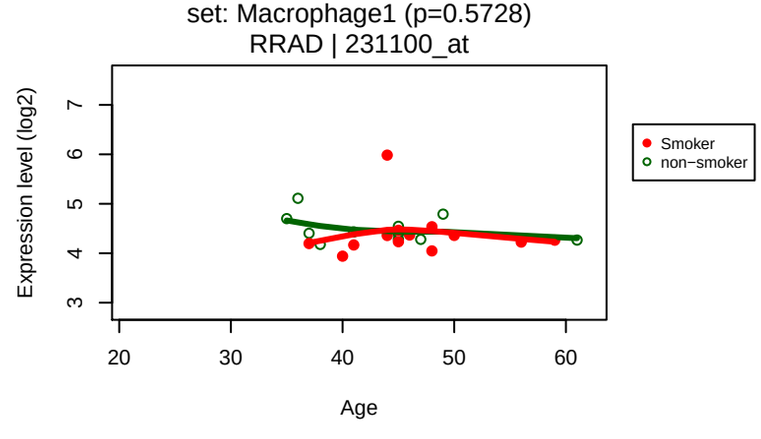

set: SmallAirway1 (p=0.2359)
RRAD | 231100_at
Expression level (log2)
Age
Smoker
non−smoker

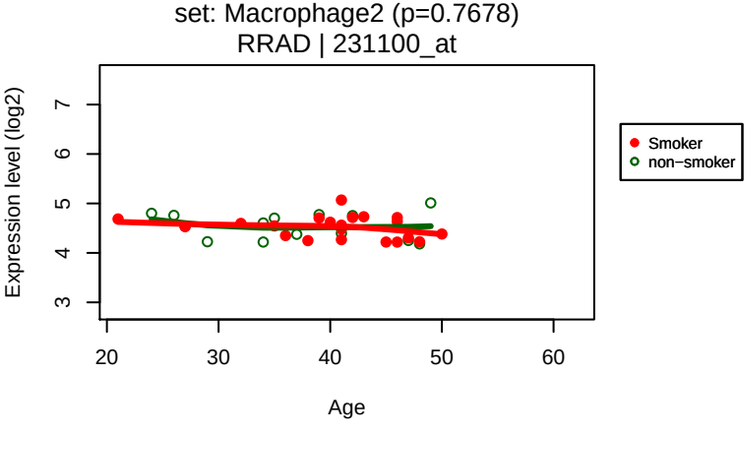

set: SmallAirway2 (p=0.3086)
RRAD | 231100_at
Expression level (log2)
Age
Smoker
non−smoker

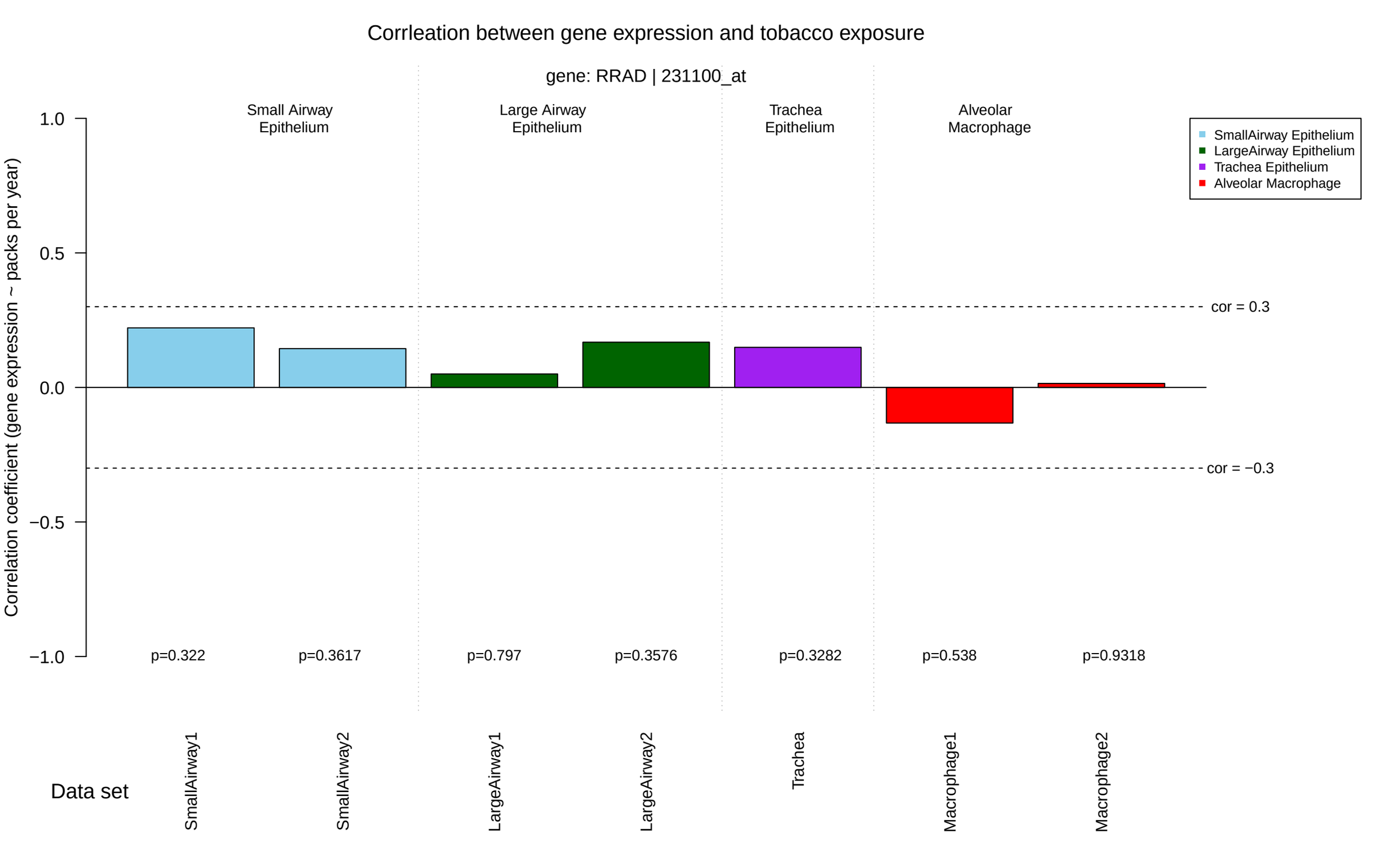

set: LargeAirway1 (p=0.5644)
RRAD | 231100_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3206)
RRAD | 231100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6979)
RRAD | 231100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5728)
RRAD | 231100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7678)
RRAD | 231100_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRAD | 231100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.322
p=0.3617
p=0.797
p=0.3576
p=0.3282
p=0.538
p=0.9318
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2